

C++ FUNDAMENTALS

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History of C++

- Developed by Bjarne Stroustrup at AT & T Bell Laboratory of Newyork in the early 1980's.
- C + Simula67 = C++



Basics

- Execution begins at main()
- C++ statements terminate with semicolons
- Comment - two types
 - //this is single comment
 - /*this is multi line comment syntax.
etc....*/
- Output Operator: cout<<
cout - (predefined object of standard output stream)
<< - insertion or put to operator
- Input Operator : cin>>
- Cin - predefined object of standard input stream
- >> - extraction operator or get from operator.



C++ program to add two numbers

```
#include<iostream>
Using namespace std;
int main()
{
float n1,n2;
cout<<"Enter two numbers : ";
cin>>n1>>n2;
float sum;
sum = n1+n2;
cout<<" Sum = " << sum;
return 0;
}
```

- Allows declaration any where in program.



Structure of C++ program

Include Files

Class Declaration

**Member function
Definitions**

Main Function Program



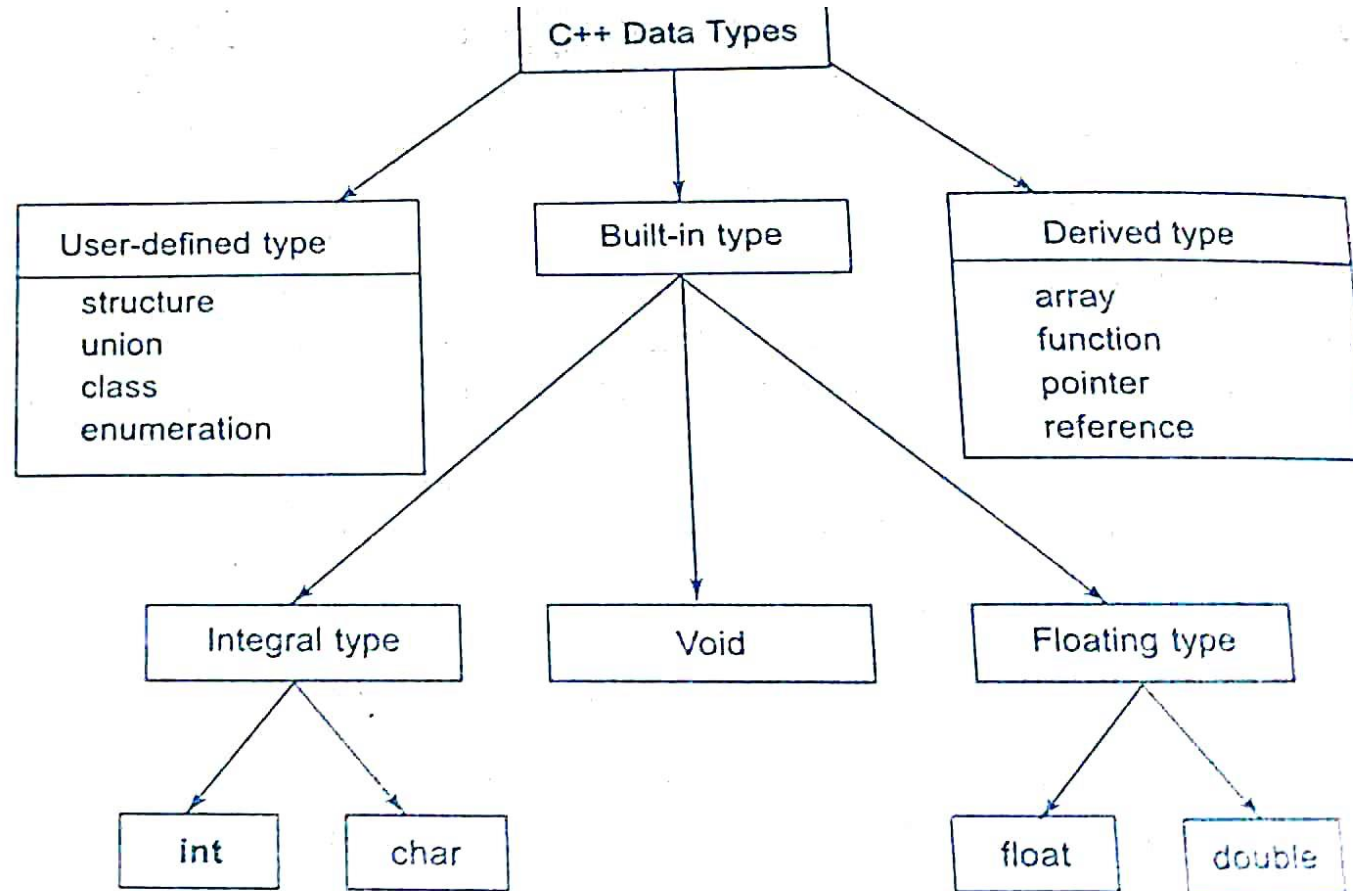
C++ Tokens

Smallest individual units

- Keywords are reserved words
- Identifiers refers to names
- Constants - fixed values that do not change during the execution of a program.
- Strings - set of characters
- Operators - symbols referring an action



Data types



Built-in Data Types

- Several of the basic types can be modified using one or more of these type modifiers –
 - signed
 - unsigned
 - short
 - long



Built-in Data Types

Type	Typical Bit Width	Typical Range
char	1byte	-127 to 127 or 0 to 255
unsigned char	1byte	0 to 255
signed char	1byte	-127 to 127
int	4bytes	-2147483648 to 2147483647
unsigned int	4bytes	0 to 4294967295
signed int	4bytes	-2147483648 to 2147483647
short int	2bytes	-32768 to 32767
unsigned short int	2bytes	0 to 65,535
signed short int	2bytes	-32768 to 32767
long int	8bytes	-2,147,483,648 to 2,147,483,647
signed long int	8bytes	same as long int
unsigned long int	8bytes	0 to 4,294,967,295
long long int	8bytes	-(2 ⁶³) to (2 ⁶³)-1
unsigned long long int	8bytes	0 to 18,446,744,073,709,551,615
float	4bytes	
double	8bytes	
long double	12bytes	
wchar_t	2 or 4 bytes	1 wide character



User defined Data type

- Structure
- Union
- Class
- Enum



Derived Data Types

- Array
- Function
- Pointer
- Reference



Reference Variable

- Provides an alias for a previously defined variable

Syntax:

data type &reference name = variable name

Example:

```
float total;
```

```
float &sum = total;
```



Operators in C++

- An operator is a symbol that tells the compiler to perform specific mathematical or logical manipulations. C++ is rich in built-in operators and provide the following types of operators –
- Arithmetic Operators – (+, -, *, /, --, ++)
- Relational Operators – (==, !=, >, <, >=, <=)
- Logical Operators – (&&, ||, !)
- Bitwise Operators – (&, |, ^, ~, <<, >>)
- Assignment Operators – (=, +=, -=, *=, /=, %=, <<=, >>=, &=, ^=, |=)
- Misc Operators – (sizeof, ?:, comma, ., pointer, address of, cast)



Additional in C++

- Scope Resolution Operator (::)
- Pointer to member declarator (::*)
- Pointer to member operator(->*)
- Pointer to member operator(.*)
- delete - memory deallocate
- new - memory allocate
- setw - field width operator



Control Structures

- Sequence
- Selection (if, if-else, switch)
- Loop (do...while, while, for)



Functions

- Block of code executed when called.
- Two types
 - Predefined
 - User defined
- Function declaration
- Function definition
- Function call



Function Example

```
#include <iostream>

using namespace std;

void swap(int a, int b) {           //here a
and b are formal parameters

    b = a + b;

    a = b - a;

    b = b - a;

    cout<<"\nAfter swapping: ";

    cout<<"a = "<<a;

    cout<<"\tb = "<<b;

    return;

}
```

```
int main()

{

    int a,b;

    cout<<"Enter the two numbers to be swapped: ";

    cin>>a>>b;

    cout<<"a = "<<a;

    cout<<"\tb = "<<b;

    swap(a,b);           //here a and b are actual parameters

}
```

Output:

Enter the two numbers to be swapped: 5 3

a = 5 b = 3

After swapping: a = 3 b = 5



Formal And Actual Parameters

- The parameters that are defined in the function definition are called **Formal Parameters**.
- The parameters in the function call which are the actual values are called **Actual Parameters**.



Passing Parameters To Functions

- Pass by Value
- Pass by Reference
- Pass by Pointer



Pass by value

```
#include <iostream>
using namespace std;
void swap(int a, int b)
{
//here a and b are formal parameters
    b = a + b;
    a = b - a;
    b = b - a;

    cout<<"\nAfter swapping inside Swap:\n ";
    cout<<"a = "<<a;
    cout<<"\tb = "<<b;
    return;
}
```

```
int main()
{
    int a,b;
    cout<<"Enter the two numbers to be
    swapped: "; cin>>a>>b;

    cout<<"a = "<<a;
    cout<<"\tb = "<<b;
    swap(a,b);
    cout<<"\nAfter swapping inside Main:\n ";
    cout<<"a = "<<a;
    cout<<"\tb = "<<b;
}
```



Pass by reference

```
#include <iostream>
#include <string>
using namespace std;
```

```
void swap(int &a, int &b)
{
    int temp = a;
    a = b;
    b = temp;
}
```

```
int main()
{
    int a,b;
    cout<<"Enter the two numbers to be swapped: ";
    cin>>a>>b;
    cout<<"a = "<<a;
    cout<<"\tb = "<<b;
    swap(a,b);
    cout<<"\nAfter swapping inside Main:\n ";
    cout<<"a = "<<a;
    cout<<"\tb = "<<b;
}
```



Pass by Pointer

```
#include <iostream>
#include <string>
using namespace std;

void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}

int main()
{
    int a,b;
    cout<<"Enter the two numbers to be swapped: "; cin>>a>>b;
    cout<<"a = "<<a;
    cout<<"\tb = "<<b;
    swap(a,b);
    cout<<"\nAfter swapping inside Main:\n ";
    cout<<"a = "<<a;
    cout<<"\tb = "<<b;
}
```

